CLASS 7 – ALGEBRA (TERM 1)

1. Explore the expression 2n+3 for different values of n.

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Create a slider by entering the following command in the Input Bar: n = Slider(0, 10, 1) and Set the initial value of n to 2.

Step 3: Enter the algebraic expression: value = 2*n + 3

Step 4: Display the expression: Text("Expression: 2n + 3", (1, 5))

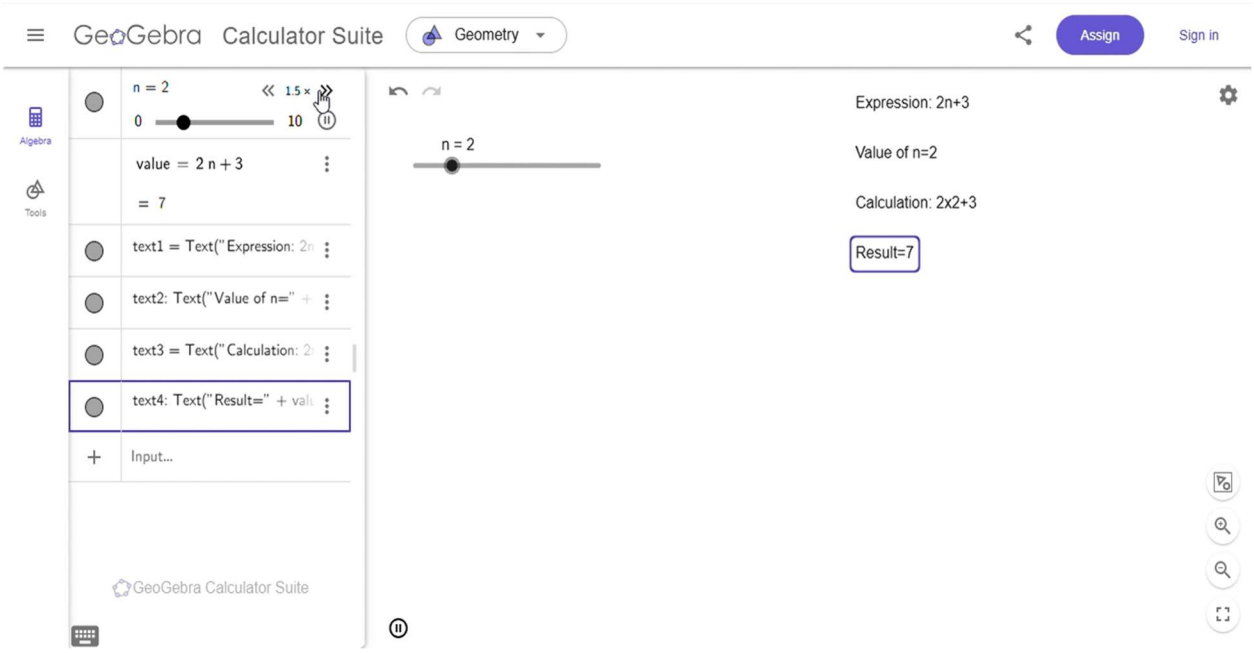
Step 5: Display the value of n: Text("Value of n = " + n, (1, 4))

Step 6: Display the calculation: Text("Calculation: 2 x " + n + " + 3", (1, 3))

Step 7: Display the result: Text("Result = " + value, (1, 2))

Step 8: Move the slider to n = 2 and observe the result. Calculation: 2(2) + 3 = 7.

Output:



2. Verify the distributive property: 3(x+2)=3x+6

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Create a slider for x by entering the following command: x=Slider(0, 10, 1) and Set the initial value to 2.

Step 3: Calculate the left-hand side by entering: L = 3*(x + 2)

Step 4: Calculate the right-hand side by entering: R = 3*x + 6

Step 5: Calculate the difference between both sides: difference = L - R

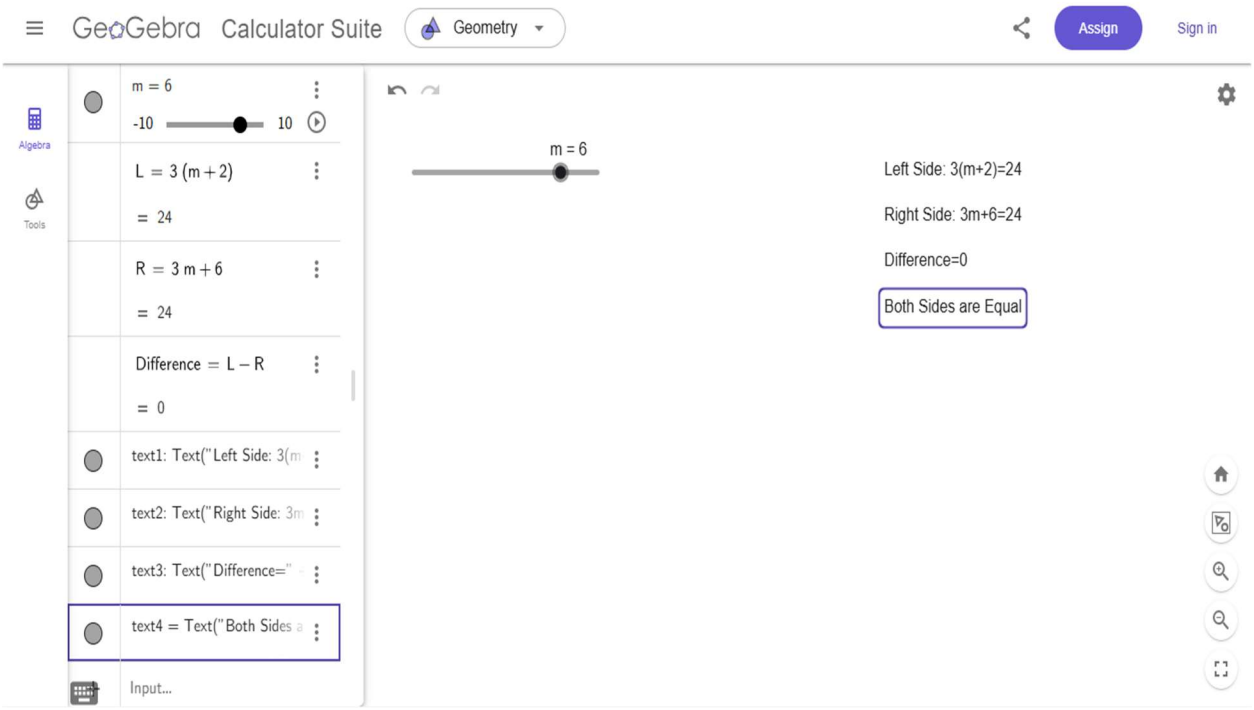
Step 6: Display the left-hand side: Text("Left side: 3(x + 2) = " + L, (1, 5))

Step 7: Display the right-hand side: Text("Right side: 3x + 6 = " + R, (1, 4))

Step 8: Display the difference: Text("Difference = " + difference, (1, 3))

Step 9: Display the conclusion:Text("Both sides are equal", (1, 2))

Output:



3. Find two Consecutive Natural numbers whose sum is 75.

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Create a slider by entering the following command in the Input Bar. a = Slider(1, 74, 1)

Step 3: Define the first number. first = a

Step 4: Define the second consecutive number. second = a + 1

Step 5: Calculate the sum of the two numbers. total = first + second

Step 6: Display the question. Text("Question: Find two consecutive natural numbers whose sum is 75.", (1, 6))

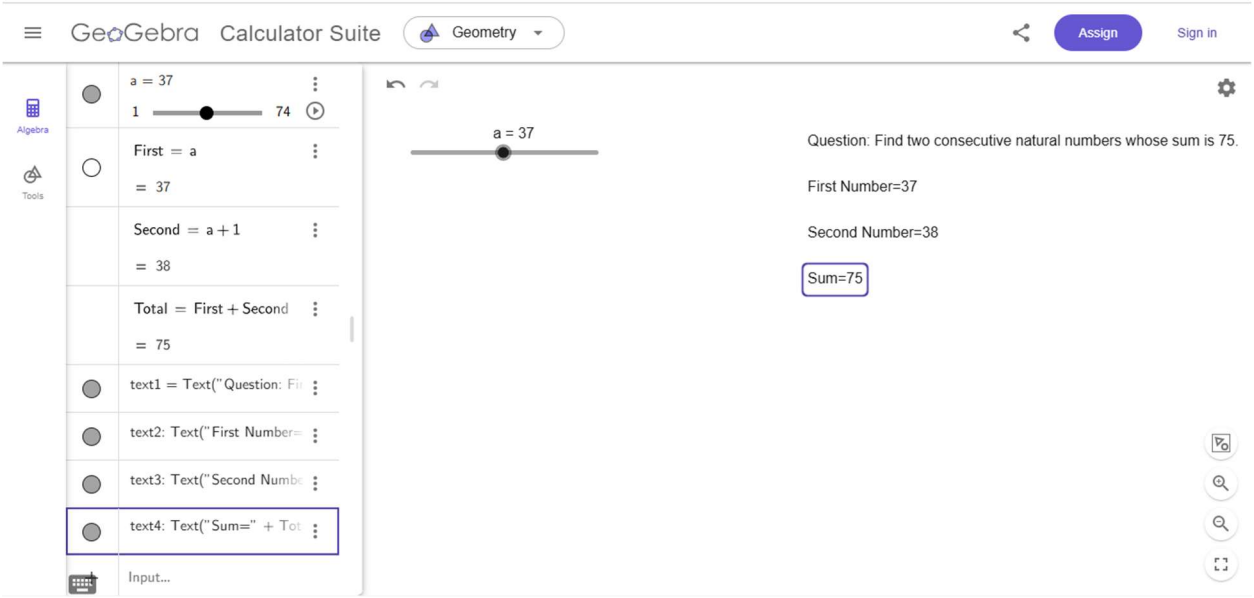
Step 7: Display the first number. Text("First number = " + first, (1, 5))

Step 8: Display the second number. Text("Second number = " + second, (1, 4))

Step 9: Display the sum. Text("Sum = " + total, (1, 3))

Step 10: Move the slider until the sum equals 75.

Output:



4. Add the following algebraic expressions: a+5b+7c and 2a+10b+9c.

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Create three sliders by entering the following commands separately.

a = Slider(0, 10, 1), b = Slider(0, 10, 1) and c = Slider(0, 10, 1)

Step 3: Enter the first expression. S1 = a + 5*b + 7*c

Step 4: Enter the second expression. S2 = 2*a + 10*b + 9*c

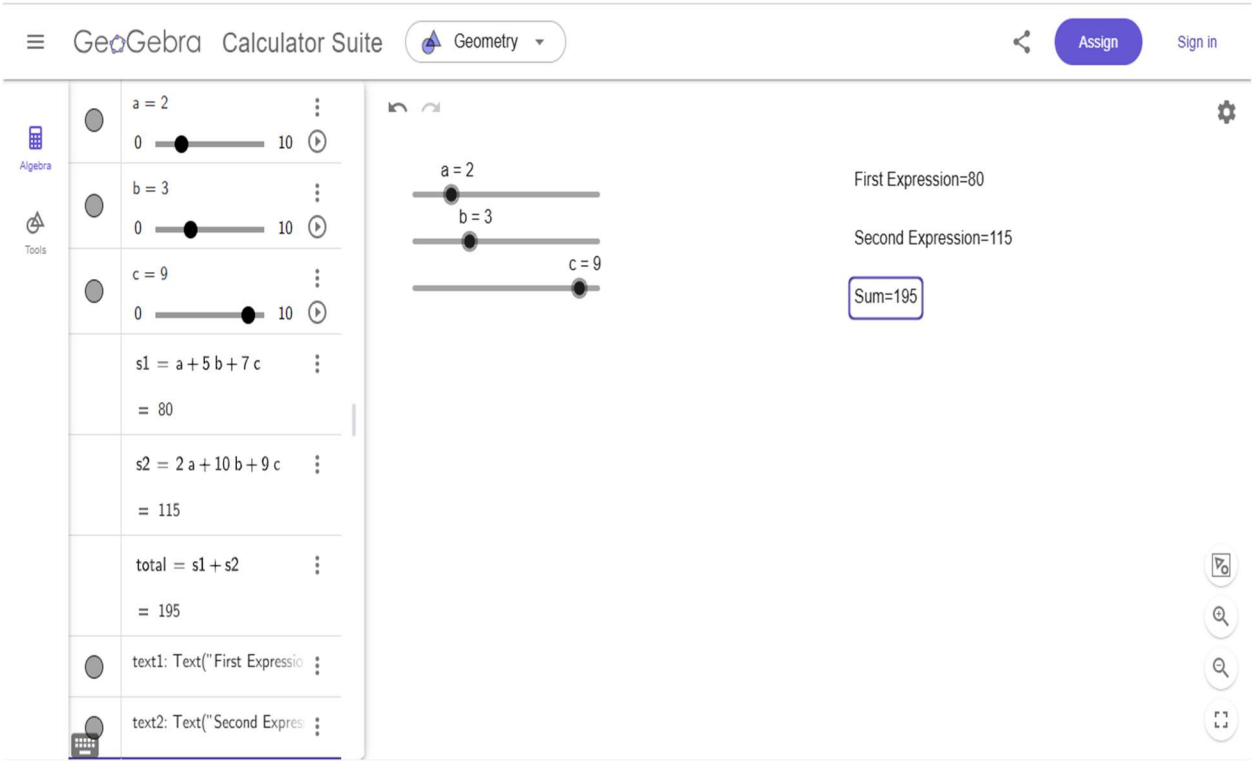
Step 5: Calculate the sum of both expressions. Total = S1 + S2

Step 6: Display the first expression. Text("First expression = " + S1, (1, 5))

Step 7: Display the second expression. Text("Second expression = " + S2, (1, 4))

Step 8: Display the sum. Text("Sum = " + Total, (1, 3))

Output:



5. Solve the following Equation: x+5=12

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Create a slider by entering the following command: x = Slider(0, 12, 1)

Step 3: Calculate the left-hand side: L = x + 5

Step 4: Define the right-hand side: R = 12

Step 5: Calculate the difference between both sides: difference = L - R

Step 6: Display the equation: Text("Equation: x + 5 = 12", (1, 5))

Step 7: Display the left-hand side: Text("Left side = " + L, (1, 4))

Step 8: Display the right-hand side: Text("Right side = " + R, (1, 3))

Step 9: Display the difference: Text("Difference = " + difference, (1, 2))

Step 10: Move the slider until x = 7. Observe that both sides equal 12 and the difference is 0.

Output:

